

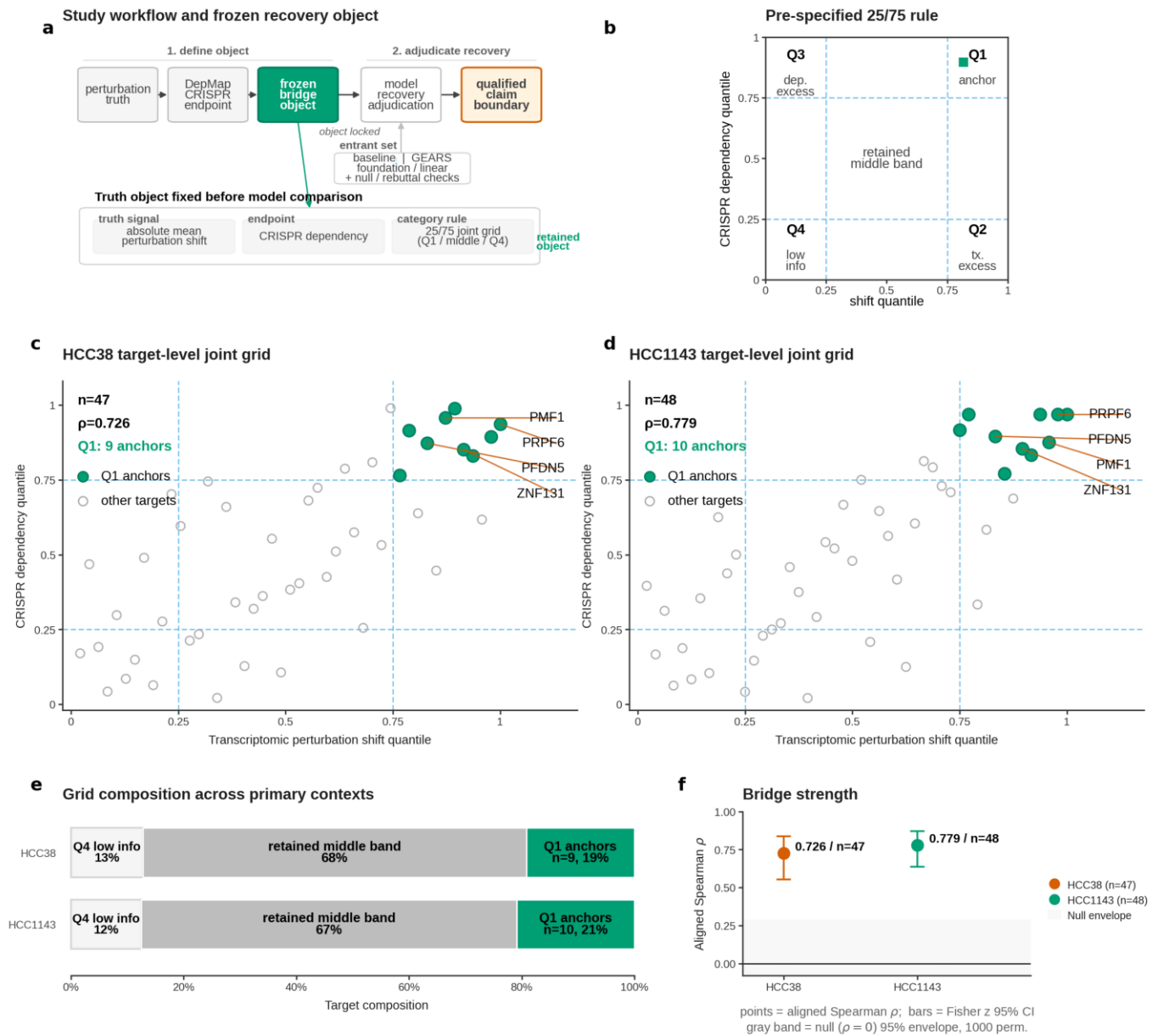


Fig. 1 | Truth object definition. a, Study workflow and frozen recovery object. b, Pre-specified 25/75 category rule. c, HCC38 target-level joint grid. d, HCC1143 target-level joint grid. e, Grid composition across the two primary contexts. f, Bridge strength summary.

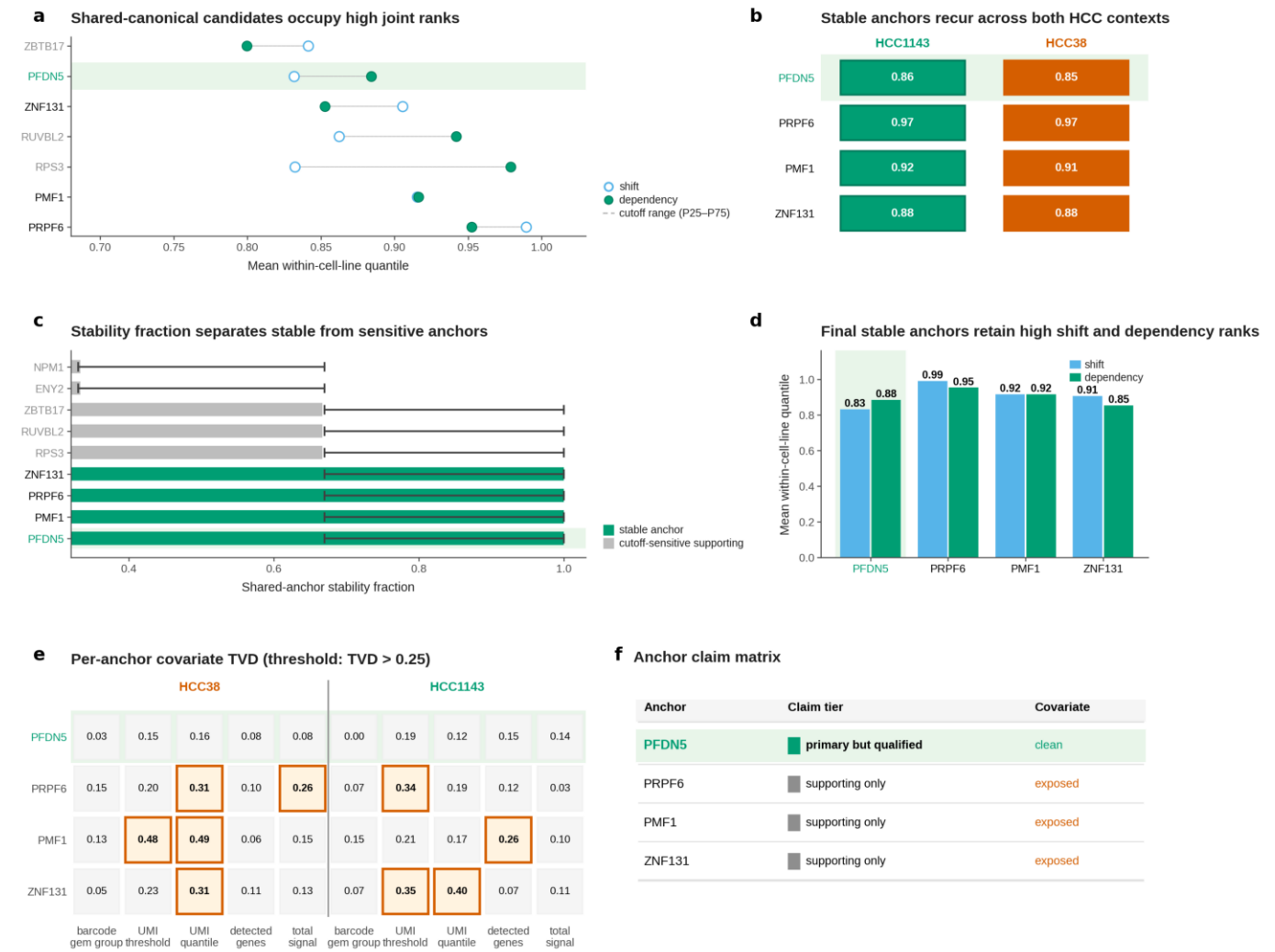


Fig. 2 | Anchor tiering. a, Shared-canonical anchor ranking. b, Low-link recurrence matrix across the two primary HCC contexts (HCC1143 and HCC38). c, Stability fraction across stable and cutoff-sensitive shared-canonical objects. d, Final stable anchors retain high shift and dependency ranks across HCC38 and HCC1143. e, Per-anchor covariate Total Variation Distance (TVD) matrix across five covariate axes. f, Compact anchor claim matrix (conclusion).

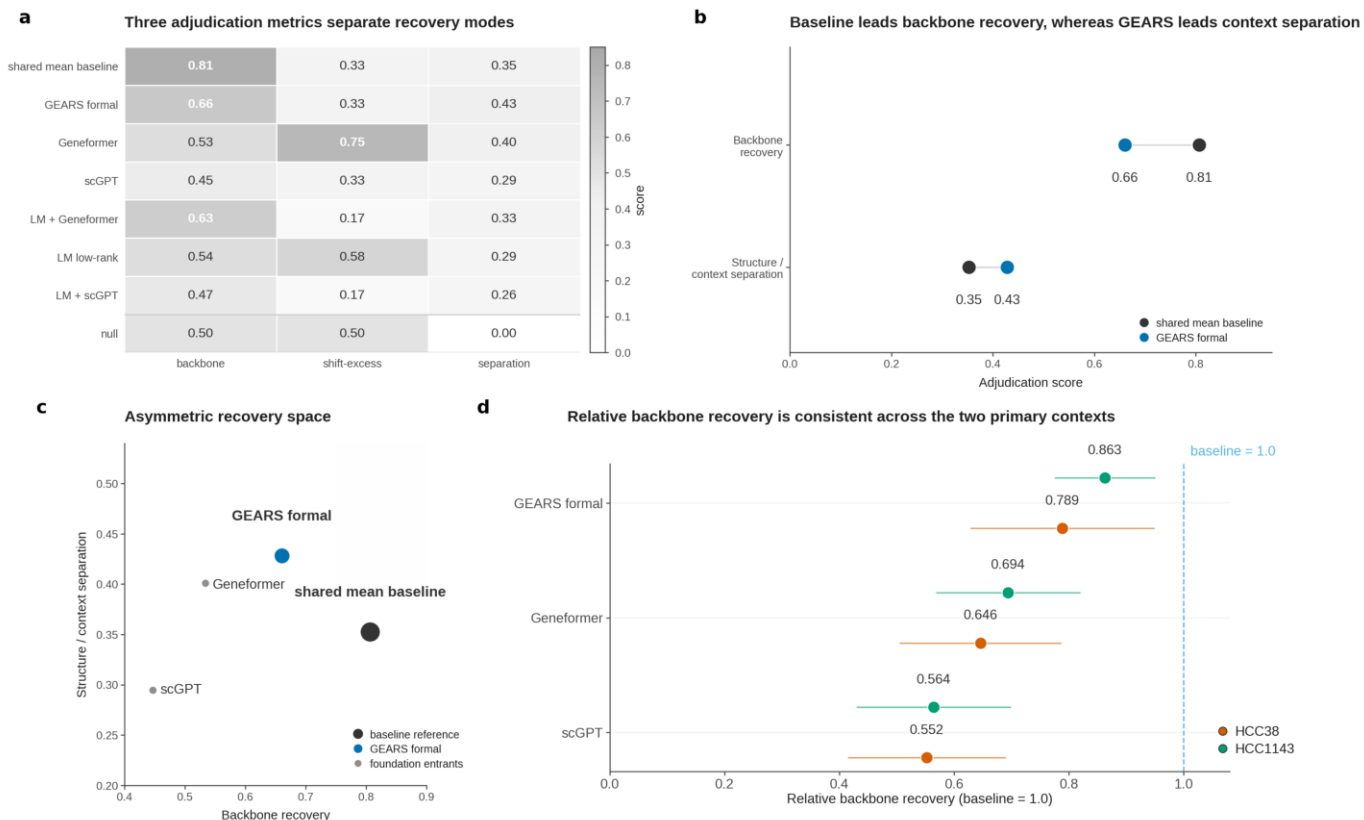
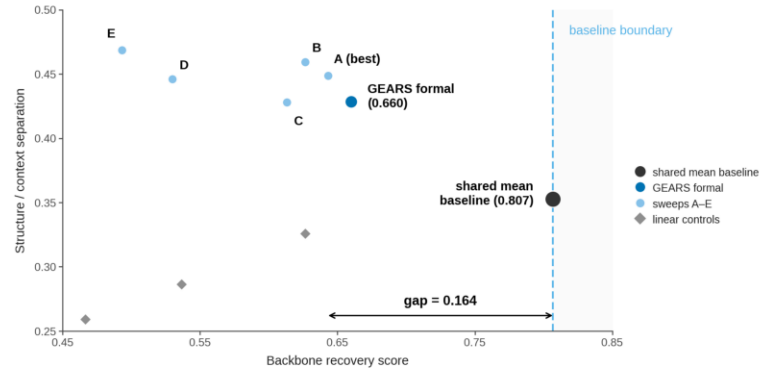


Fig. 3 | Model adjudication. a, Three adjudication metrics separate recovery modes across entrants. b, Baseline leads backbone recovery, whereas GEARS leads context separation. c, Entrants occupy an asymmetric recovery space.

a Pre-specified finite-budget GEARS candidates

Candidate	Epoch	LR	WD	Role
GEARS formal	30	1×10^{-3}	1×10^{-6}	reference recipe
A (best)	30	2×10^{-3}	1×10^{-6}	best sweep candidate
B	20	1×10^{-3}	1×10^{-6}	sweep candidate
C	30	1×10^{-3}	1×10^{-5}	sweep candidate
D	30	5×10^{-4}	1×10^{-6}	sweep candidate
E	40	1×10^{-3}	1×10^{-6}	sweep candidate

b Prespecified rebuttal candidates do not close the backbone gap



c No tested rebuttal candidate closes the backbone gap to the shared-mean baseline

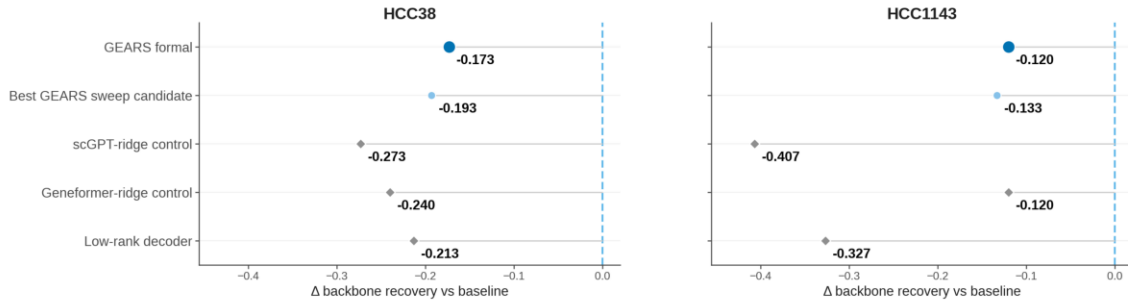


Fig. 4 | Finite-budget rebuttal tests. a, Pre-specified finite-budget GEARS candidates. b, Pre-specified rebuttal recovery space. c, Per-context residual backbone gap.

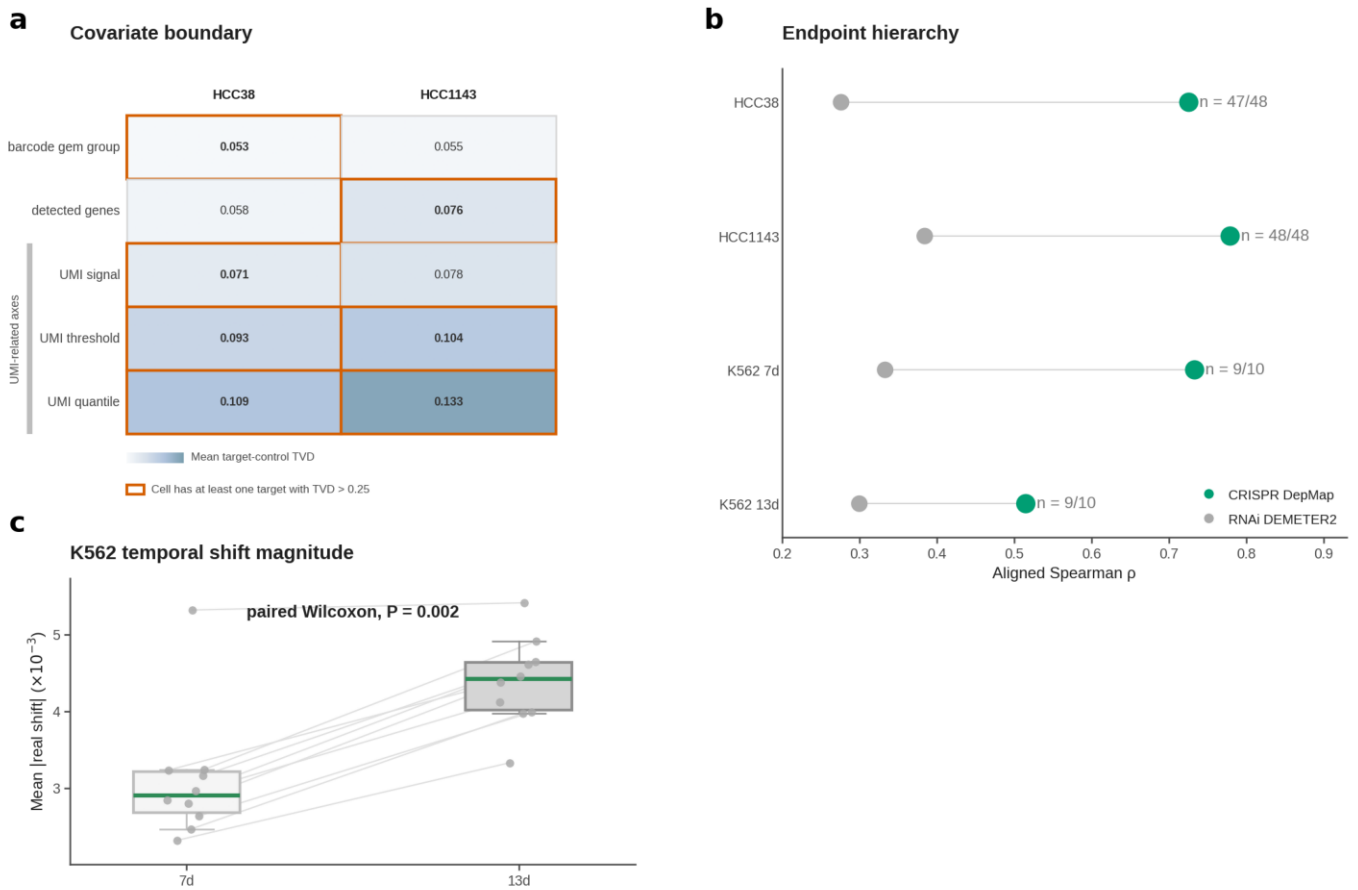
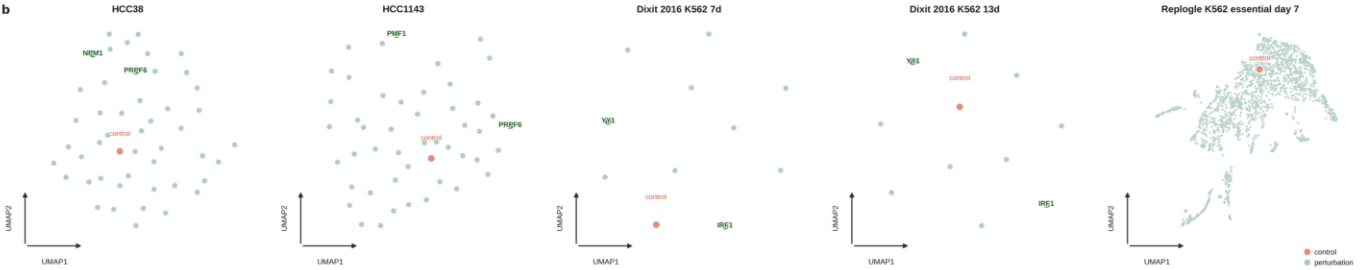


Fig. 5 | Boundary governance. a, Covariate boundary in HCC38 and HCC1143 (Framework element 7). b, Endpoint hierarchy across all four bridge contexts (Framework element 6). c, K562 temporal stratification: stronger rank alignment at 7d despite smaller perturbation magnitude. d, A0/A1/B evidence-tier summary for K562 temporal evidence (Framework element 6).

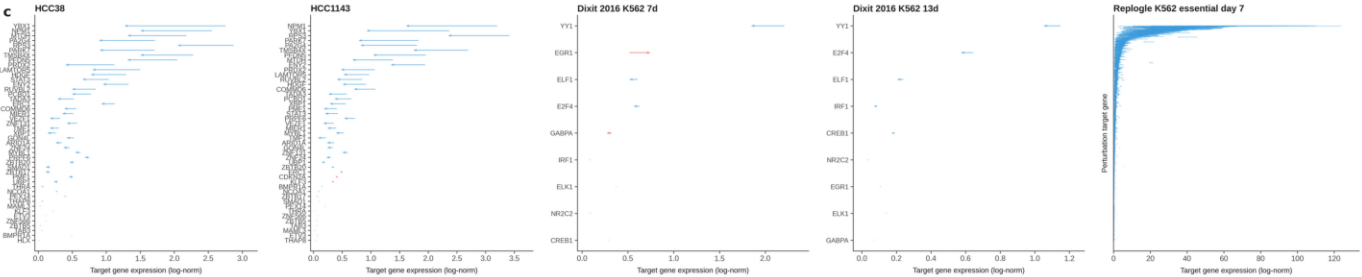
a

Dataset	Size	Benchmark use
Perturbation datasets		
HCC38	36,601 genes x 14,175 cells	47 targets 1,666 controls
HCC1143	36,601 genes x 11,405 cells	47 targets 1,325 controls
Dixit 2016 K562 7d	20,824 genes x 28,034 cells	10 targets 5,381 controls
Dixit 2016 K562 13d	20,824 genes x 15,849 cells	10 targets 3,491 controls
Replogle K562 essential day 7	8,563 genes x 310,385 cells	2057 targets 10,691 controls
Endpoint datasets		
DepMap CRISPR dependency (primary endpoint)	1,186 cell lines x 18,435 genes	HCC38 47 / HCC1143 48 / K562 7d 10 / K562 13d 10 / Replogle K562 1,882
DEMETER2 RNAi (sensitivity endpoint)	701 mapped cell lines x 17,309 genes	HCC38 48 / HCC1143 48 / K562 7d 9 / K562 13d 9 / Replogle K562 1,882

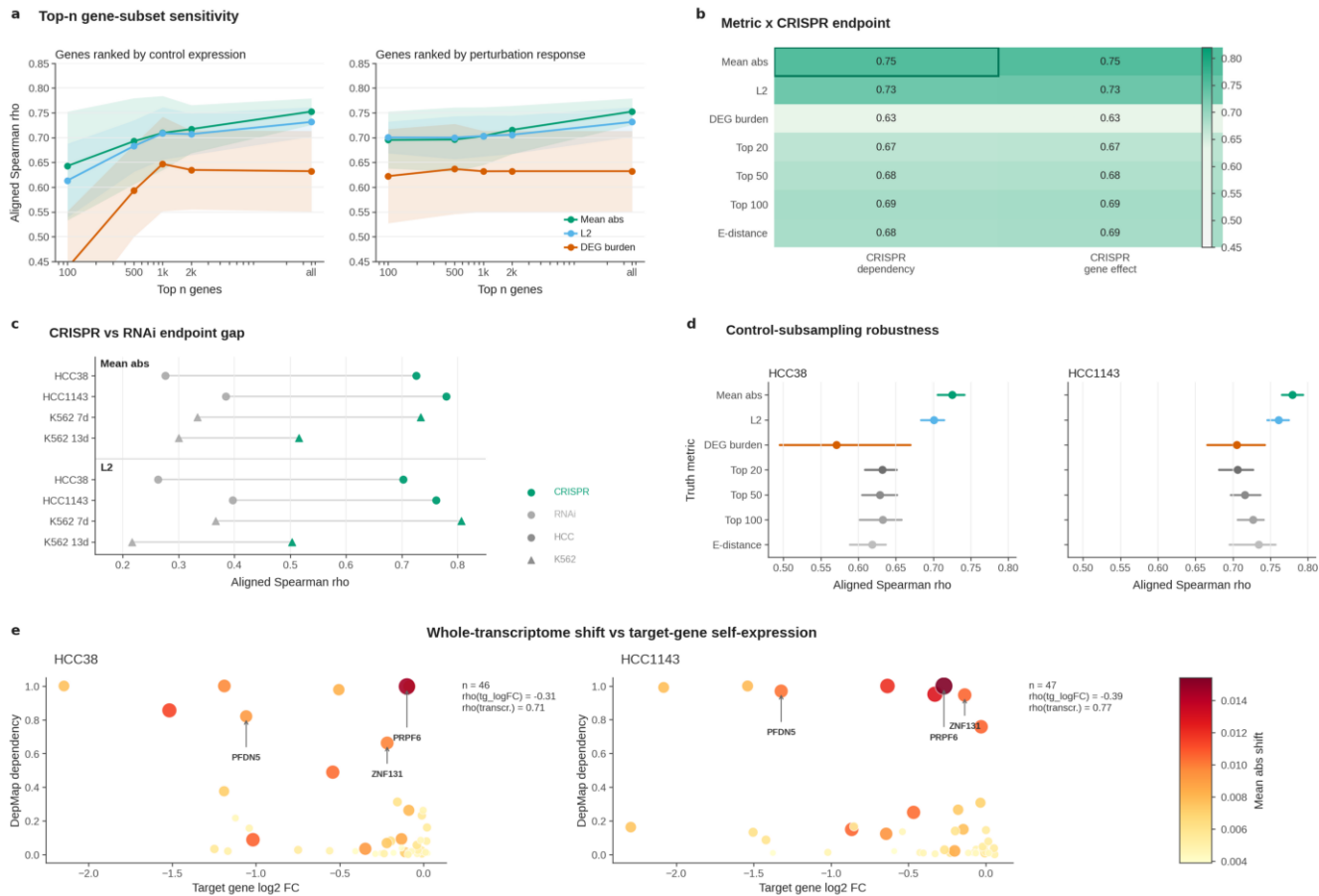
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c

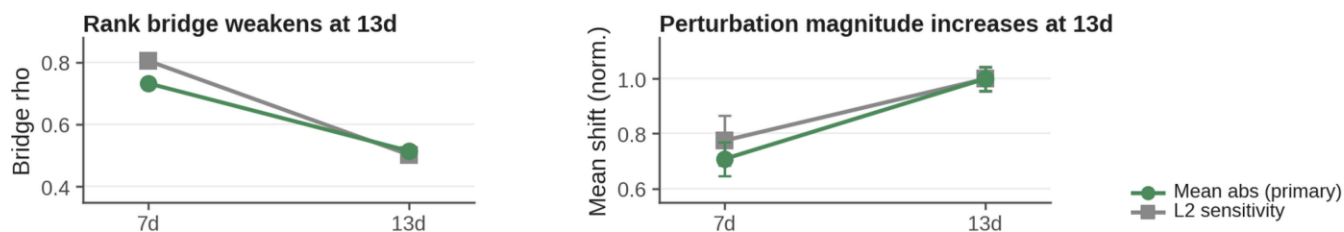


Extended Data Fig. 1 | Dataset familiarization. a, Dataset overview table. b, UMAP of perturbation-level mean profiles for HCC38, HCC1143, Dixit 2016 K562 7d, Dixit 2016. c, Target-gene expression change for the same five contexts (left to right).

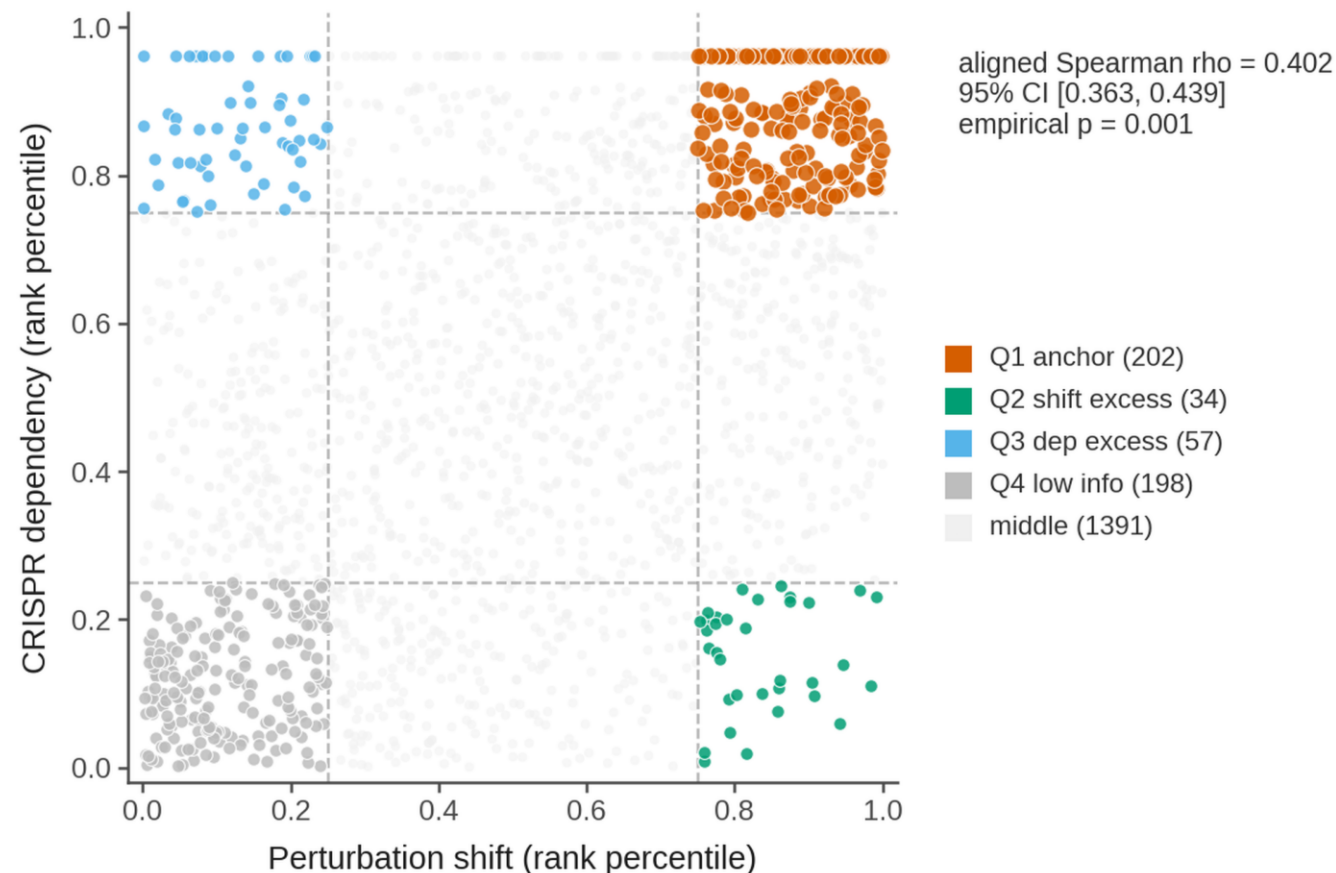


Extended Data Fig. 2 | Metric robustness. a, Top-n gene-subset sensitivity. b, Metric and CRISPR endpoint robustness. c, Endpoint sensitivity. d, Control-subsampling robustness. e, Whole-transcriptome shift versus target-gene self-expression in driving the fitness bridge.

a K562 temporal stratification (7d vs 13d)

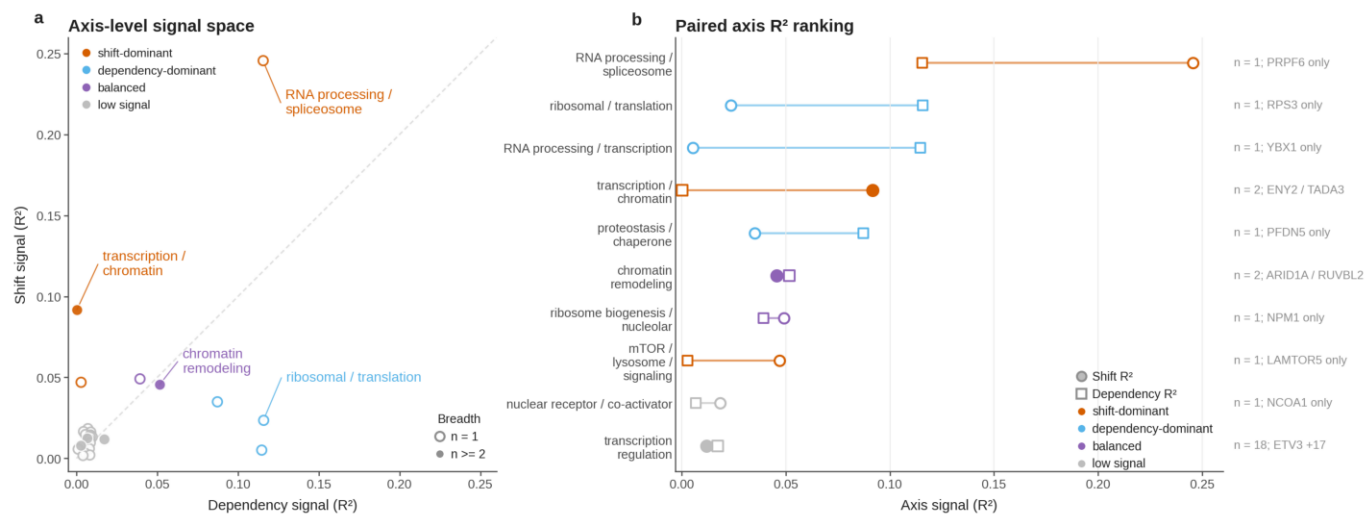


b Large-scale perturbation-fitness bridge confirmation



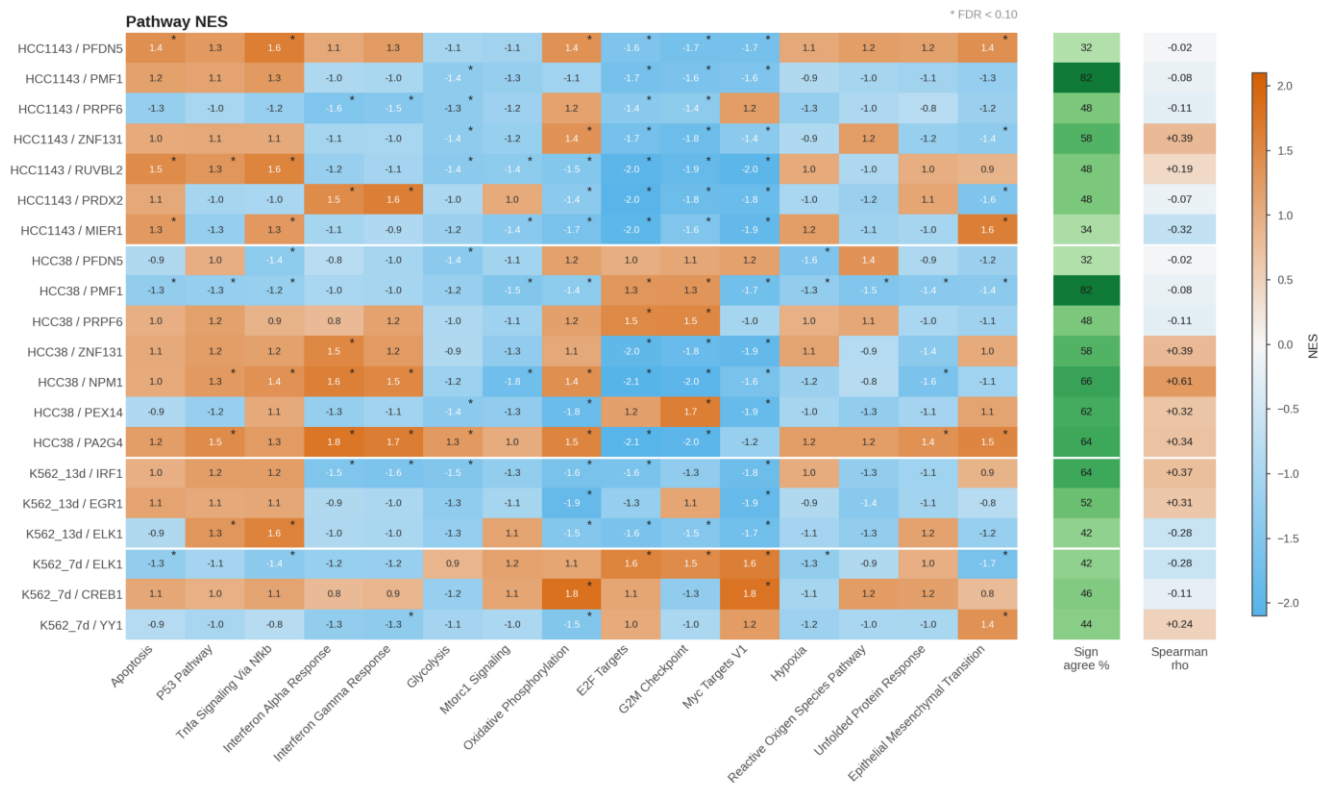
Extended Data Fig. 3 | K562 and Replogle bridge support.

a, K562 temporal bridge-magnitude dissociation (GSE90063). b, Replogle K562 essential day 7 large-scale perturbation-fitness bridge.



Extended Data Fig. 4 | Axis-level signal space. a, Axis-level signal space comparing dependency signal with transcriptomic shift signal across annotated axes. b, Paired axis R-squared ranking.

a Exploratory pathway-level summaries of target-control perturbation responses



Extended Data Fig. 5 | Pathway-response polarity. a, Pathway normalized enrichment score (NES) heat map across the pre-specified display targets in HCC38, HCC1143, K562 7d, and K562 13d.